

Minki Lee

CONTACT INFORMATION	Department of Mathematics University of Michigan 530 Church St, Ann Arbor, MI 48109	Homepage: minkiplee.github.io ✉ E-mail: minkilee@umich.edu Tel: +1 734-369-0435
EDUCATION	Columbia University, New York, NY M.D./Ph.D., Medical Scientist Training Program (MSTP) University of Michigan, Ann Arbor, MI Master of Science in Mathematics University of Michigan, Ann Arbor, MI Bachelor of Science in Mathematics	Aug 2026 – 2034 (Expected) Sep 2023 – Dec 2024 Sep 2020 – Dec 2024
RESEARCH EXPERIENCES	University of Michigan, Ann Arbor, MI Advisor: Dr. Daniel Forger Institute for Basic Science, Daejeon, South Korea Advisor: Dr. Jae Kyoung Kim	Sep 2021 – present May 2022 – Aug 2022
RESEARCH INTERESTS	Computational Psychiatry, Biomedical Data Science, Circadian Rhythms, Sleep, Mobile Health, Machine Learning, Nonlinear Dynamics	
PUBLICATIONS	* : co-first author, † : co-corresponding author Kim R*, Fang Y, Lee MP , Kim DW, Tang Z, Sen S, Forger DB†, Seasonal timing and interindividual differences in shiftwork adaptation, <i>NPJ Digit. Med.</i> (2025). Mayer C*, Kim DW*, Zhang M, Lee MP , Forger DB, Burgess HJ, Moon C†, Predicting circadian phase in community-dwelling later-life adults using actigraphy data, <i>J. Sleep Res.</i> (2024). Lee MP* , Kim DW*,†, Fang Y, Kim R, Bohnert ASB, Sen S, Forger DB†, The real-world association between digital markers of circadian disruption and mental health risks, <i>NPJ Digit. Med.</i> (2024). Lee MP* , Kim DW*,†, Mayer C, Walch O, Forger DB, The combination of topological data analysis and mathematical modeling improves sleep stage prediction from consumer-grade wearables, <i>J. Biol. Rhythms</i> (2024). Lee MP* , Hoang K*, Park S, Song YM, Joo EY, Chang W†, Kim JH†, Kim JK†, Imputing missing sleep data from wearables with neural network in real-world settings, <i>Sleep</i> (2023). Kim DW*, Lee MP* , Forger DB†, Wearable data assimilation to estimate the circadian phase, <i>SIAM J. Appl. Math</i> (2023). Kim DW*,†, Mayer C*, Lee MP , Choi SW, Tewari M, Forger DB, Efficient assessment of real-world dynamics of circadian rhythms in heart rate and body temperature from wearable data, <i>J. R. Soc. Interface</i> (2023).	
HONORS AND AWARDS	2025 Student Travel Grant 2024 Best Poster Presentation Award 2024 Rackham Graduate School International Travel Grant 2024-2025 Outstanding Achievement in Mathematics Award 2023 Barry M. Goldwater Scholarship 2023 Wilfred Kaplan Award in Applied Mathematics 2020-2024 Department of Mathematics Merit Scholarship 2020 Andrew J. Lum & David R. Juillet Scholarship	National Institute for Theory and Mathematics in Biology Society for Mathematical Biology University of Michigan University of Michigan Goldwater Foundation University of Michigan University of Michigan Ann Arbor Community Foundation

PRESENTATION	2025 NSF-Simons Foundation NITMB MathBio Convergence Conference, Poster, Chicago, IL
	2025 Biological Research Information Center Seminar, POSTECH, Online
	2024 Society for Mathematical Biology Annual Meeting, Poster, Seoul, South Korea
	2024 Society for Research on Biological Rhythms Biennial Meeting, Poster, San Juan, Puerto Rico
	2023 SIAM Great Lakes Section Annual Meeting, Contributed talk, Lansing, MI, USA
	2022 SIAM Great Lakes Section Annual Meeting, Minisymposium, Detroit, MI, USA
	2022 IBS Biomedical Mathematics Seminar, Daejeon, South Korea
	2022 Annual Conference of Korean Society for Industrial and Applied Mathematics, Poster, Daejeon, South Korea
	2021 University of Michigan Mathematics REU Seminar, Virtual
PEER REVIEW ACTIVITY	npj Digital Medicine, npj Biosensing, Scientific Reports